

Topology Optimization

Lecture 2 – FEM 1

Poisson's Equation

The Finite Element Method

Properties of the Finite Element Method

Course overview

No.	Topics
1	Introduction Function and vector spaces, Interpolation, L^2 projection
2	The Finite Element Method for PDEs 1: Poisson's Equation, The Finite Element Method, Properties of the Finite Element Method
3	The Finite Element Method for PDEs 2: Index notation, Linear Elasticity, Non-linear Poisson's equation, Non-linear Elasticity
4	Constrained Optimization 1: Optimality, KKT Theorem, Numerical Optimization Algorithms
5	Constrained Optimization 2: Method of Moving Asymptotes, MPI and PETSc
6	Introduction to Topology Optimization: The idea, Filtering and Projection, Python implementation
7	Gradients
8	Robust topology optimization, More on MPI
9	Wave equations
10	Acoustic waves and more on Robust Optimization
11	Penalty and Augmented Lagrangian Methods Stress-constraints in Topology Optimization 1
12	Stress-constraints in Topology Optimization 2
13	Light concentration using nanoparticle design
14	Heat conduction

Some mathematical prerequisites ✓ **DONE!**

Mathematical and numerical
formulation of the
Finite Element Method

Optimization Theory and
numerical optimization methods

Basic Topology Optimization

Advanced Topology Optimization

Applications of Topology Optimization

The Finite Element Method

- We will today go through the basics of how to solve Partial Differential Equations using the finite element method.
- We will look at the '[Hello World](#)' example of FEM, namely Poisson's equation.
- This will be converted to the weak form, and we will use the knowledge of function spaces from last week to derive a set of linear equations.
- When solving these equations, we obtain the numerical solution to the PDE.
- We will also look at some of the mathematical properties of the finite element method.
- At the exercises:
 - You will implement a solver in python/FEniCS by following Chapter 2 in "Solving PDEs in Python".
 - You will go through some of the theory on these slides yourself in Exercise 2.

Poisson's Equation

In the finite element method, the PDEs are reformulated as integral equations, the so-called weak form.

We consider here Poisson's Eq. for finding the electric potential in electrostatics. Poisson's equation also appear in many other applications, e.g., in heat transfer, fracture mechanics, fluid flow, thin plates etc.

Maxwell's equations are

$$\begin{array}{ccccccc} \text{Gauss' law} & & \text{Faraday's law} & & \text{Gauss' law for} & & \text{Ampere's law} \\ & & \text{of induction} & & \text{the magnetic field} & & \text{(with Maxwell correction)} \\ \nabla \cdot \vec{E} = \frac{\rho}{\epsilon} & , & \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} & , & \nabla \cdot \vec{B} = 0 & , & \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \end{array}$$

For static cases, $\frac{\partial}{\partial t} = 0$ leading to the definition of the voltage (electric potential) u as

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = 0 \Leftrightarrow \vec{E} = -\nabla u \text{ since } \nabla \times \nabla u = 0 \text{ for any scalar field } u.$$

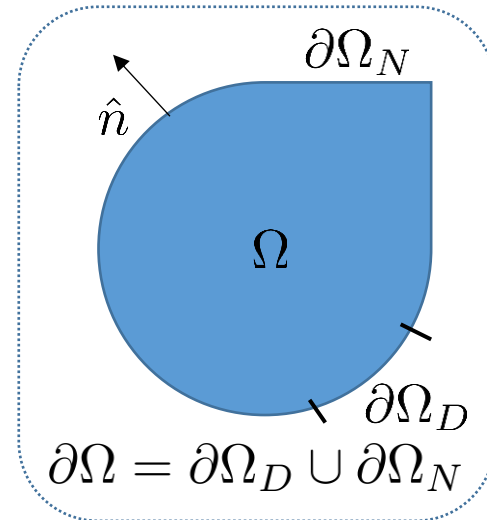
Inserting into Gauss' law yield

$$\nabla \cdot \vec{E} = -\nabla^2 u = -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} - \frac{\partial^2 u}{\partial z^2} = \frac{\rho}{\epsilon} \equiv f \quad \text{in } \Omega$$

The PDE is supplemented by boundary conditions:

Dirichlet, voltage is specified: $u = u_0 \quad \text{on } \partial\Omega_D \subset \partial\Omega$

Neumann, normal component of $\vec{E}$ is specified: $\nabla u \cdot \hat{n} = g \quad \text{on } \partial\Omega_N \subset \partial\Omega$



Poisson: $-\nabla^2 u = f \text{ in } \Omega$
 $u = u_0 \text{ on } \partial\Omega_D$
 $\nabla u \cdot \hat{n} = g \text{ on } \partial\Omega_N$

Solving the PDE with the given boundary conditions yield the voltage. From the voltage, the electric field can be calculated.

Both are important quantities in electrostatics.

Weak Form

Poisson: $-\nabla^2 u = f$ in Ω
 $u = u_0$ on $\partial\Omega_D$
 $\nabla u \cdot \hat{n} = g$ on $\partial\Omega_N$

Recall your favourite integral theorem, the divergence theorem:

$$\int_{\Omega} \nabla \cdot \vec{F} \, dV = \oint_{\partial\Omega} \vec{F} \cdot \hat{n} \, dS \quad , \quad \hat{n} \text{ is the outward unit normal to } \partial\Omega$$

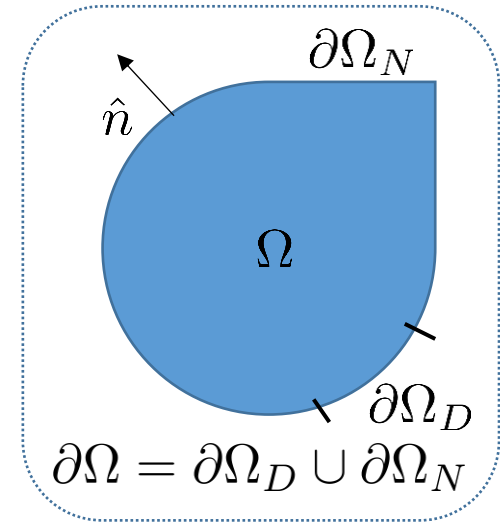
The weak form is obtained by multiplying the PDE by an arbitrary test function, v , and integrating.

We multiply the Poisson equation by an (almost) arbitrary function v and integrate:

$$\begin{aligned} \int_{\Omega} v (-\nabla^2 u) \, dV &= - \int_{\Omega} v \nabla \cdot \nabla u \, dV \\ &= - \int_{\Omega} (\nabla \cdot (v \nabla u) - \nabla v \cdot \nabla u) \, dV \\ &= - \oint_{\partial\Omega} v \nabla u \cdot \hat{n} \, dS + \int_{\Omega} \nabla v \cdot \nabla u \, dV \\ &= \int_{\Omega} v f \, dV. \end{aligned}$$

Product rule

$$\begin{aligned} \nabla \cdot (v \vec{F}) &= \nabla v \cdot \vec{F} + v \nabla \cdot \vec{F} \Leftrightarrow \\ v \nabla \cdot \vec{F} &= \nabla \cdot (v \vec{F}) - \nabla v \cdot \vec{F} \end{aligned}$$



Rearranging terms and using the boundary conditions, the weak form of Poisson's equation is obtained as

$$\int_{\Omega} \nabla u \cdot \nabla v \, dV = \int_{\partial\Omega_N} g v \, dS + \int_{\Omega} v f \, dV.$$

v is chosen to be zero on the Dirichlet part of the boundary (see next few slide), and the boundary integral reduces to only the Neumann part.

Weak Form

Classical (strong) form:

$$\begin{aligned} -\nabla^2 u &= f && \text{in } \Omega \quad , \\ u &= u_0 && \text{on } \partial\Omega_D \subset \partial\Omega \quad , \\ \nabla u \cdot \hat{n} &= g && \text{on } \partial\Omega_N \subset \partial\Omega \quad , \quad \partial\Omega_N = \partial\Omega \setminus \partial\Omega_D \end{aligned}$$

Weak form (or variational form): Find $u \in T$ such that

$$\int_{\Omega} \nabla u \cdot \nabla v \, dV = \int_{\partial\Omega_N} g v \, dS + \int_{\Omega} v f \, dV \quad , \quad \forall v \in \hat{T}$$

The Neumann boundary conditions appear directly in the weak equation.

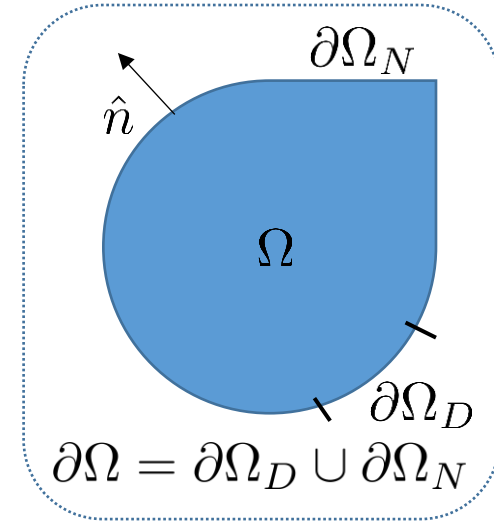
The Dirichlet boundary conditions are incorporated into the function space T .

The test function v is required to be zero on $\partial\Omega_D$, which is incorporated into the function space $\hat{T}$.

The latter two requirements make sure there are no degrees of freedom associated with the part of the geometry where the solution is already specified by the Dirichlet boundary conditions.

The weak form have some advantages:

1. Need only to integrate ∇u , not differentiate u twice as in the strong form. For ex. u can be piecewise differentiable.
2. f can be a generalized function (distribution), like the Dirac δ function, since it only needs to be integrated.
3. Integrals are easy to do numerically, even for complicated geometries, differentiation is not.



Discretization of The Weak Form

Let V be the space of functions where the integrals in the weak form "make sense":

$$V = \{v \mid \|v\|_{L^2} + \|\nabla v\|_{L^2} < \infty\} \quad , \quad \|v\|_{L^2} = \sqrt{\int_{\Omega} |v|^2 dV}$$

The weak form should be satisfied for any v in the *test space*

$$\hat{T} = \{v \in V \mid v = 0 \text{ on } \partial\Omega_D\}$$

The solution u is found in the *trial space*

$$T = \{u \in V \mid u = u_0 \text{ on } \partial\Omega_D\}$$

where the Dirichlet boundary conditions $v = 0$ and $u = u_0$ on $\partial\Omega_D$ have been used directly in the definitions of $\hat{T}$ and T , respectively.

The discretized version of the weak equation is found by restricting the test space and trial space to finite dimensional subsets of $\hat{T}$ and T :

$$\hat{T}_h \subset \hat{T} \text{ and } T_h \subset T$$

The discrete problem is: Find $u_h \in T_h$ such that

$$\int_{\Omega} \nabla u_h \cdot \nabla v_h dV = \int_{\partial\Omega_N} g v_h dS + \int_{\Omega} v_h f dV \quad , \quad \forall v_h \in \hat{T}_h$$

Discretization of The Weak Form

We can now expand the trial function in a basis of the trial space, T_h :

$$u_h = \sum_{j=1}^{n_p} U_j \phi_j$$

where ϕ_j are the basis functions ($\sim$ basis vectors) for T_h .

The test function can also be expanded like this, but it is not needed to form the discrete equations.

Inserting u_h into the weak form and using the requirement that it should hold for all $v_h \in \hat{T}_h$ (including each $\hat{\phi}_i$), we get

$$\sum_{j=1}^{n_p} U_j \int_{\Omega} \nabla \phi_j \cdot \nabla \hat{\phi}_i \, dV = \int_{\partial\Omega_N} \hat{\phi}_i g \, dS + \int_{\Omega} f \hat{\phi}_i \, dV \quad , \quad i = 1, \dots, n_p$$

Note that ϕ_j and $\hat{\phi}_i$ depends on the spatial coordinates, whereas U_j does not.

This can be written as a system of linear equations for the unknown U_j 's (put into the U vector):

$$KU = L$$
$$K_{ij} = \int_{\Omega} \nabla \phi_j \cdot \nabla \hat{\phi}_i \, dV \quad , \quad L_i = \int_{\partial\Omega_N} \hat{\phi}_i g \, dS + \int_{\Omega} f \hat{\phi}_i \, dV$$

Find $u_h \in T_h$ such that

$$\int_{\Omega} \nabla u_h \cdot \nabla v_h \, dV = \int_{\partial\Omega_N} g v_h \, dS + \int_{\Omega} v_h f \, dV \quad , \quad \forall v_h \in \hat{T}_h$$

Global vs. Local

- In the form shown to the right, the integrals are over the whole space.

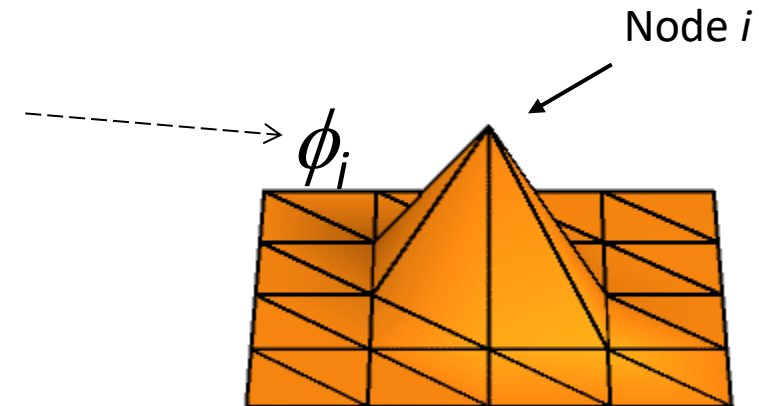
$$KU = L$$
$$K_{ij} = \int_{\Omega} \nabla \phi_j \cdot \nabla \hat{\phi}_i \, dV \quad , \quad L_i = \int_{\partial\Omega_N} \hat{\phi}_i g \, dS + \int_{\Omega} f \hat{\phi}_i \, dV$$

- Generally, all basis function will couple to all other basis functions in K .
- This makes K a *dense matrix* in the general case.
- The integrals can be split into many “smaller” integrals by linearity of integration:

$$\int_{\Omega} F \, dV = \int_{\Omega_1} F \, dV + \int_{\Omega_2} F \, dV + \int_{\Omega_3} F \, dV + \cdots + \int_{\Omega_N} F \, dV$$

with $\Omega = \bigcup_{i=1}^{n_t} \Omega_i$ and F being any integrand, e.g., $F = \nabla \phi_j \cdot \nabla \hat{\phi}_i$.

- If the basis functions are made by stitching *localized* functions together, most of these integrals will be zero and K becomes a *sparse matrix*.



Dirichlet Boundary Conditions

Let's try to find a solution from the function space P_1 :

$$T_h = \{u_h \in P_1 \mid u_h = \phi_0 \text{ on } \partial\Omega_D\}$$

$$\hat{T}_h = \{v_h \in P_1 \mid v_h = 0 \text{ on } \partial\Omega_D\}$$

The system of equations are

$$KU = L$$

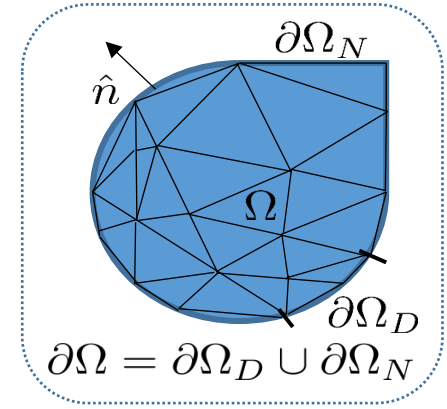
$$K_{ij} = \int_{\Omega} \nabla \phi_j \cdot \nabla \hat{\phi}_i \, dV \quad , \quad L_i = \int_{\partial\Omega_N} \hat{\phi}_i g \, dS + \int_{\Omega} f \hat{\phi}_i \, dV$$

where ϕ_i and $\hat{\phi}_i$ are the Hat functions associated with the trial and test spaces T_h and $\hat{T}_h$.

U is the vector of unknown degrees of freedom, i.e. nodal values for P_1 .

Since we now use P_1 , we will in the following equate the degrees of freedom with nodal values. This may not be the case for other function spaces.

Since the function spaces for the test and trial functions are not exactly the same, this stiffness matrix will not be symmetric!



Dirichlet Boundary Conditions

K_{ij} can be made symmetric by noting that we can write

$$u_h = u_{h,0} + u_{h,D}$$

where $u_{h,0} \in \hat{T}_h$ and $u_{h,D} \in T_h$ and $u_h \in T_h$ with

$$u_{h,Dj} = \begin{cases} u_{0j} & \text{if } j \in \text{Fixed} \\ 0 & \text{for all other nodes in the mesh} \end{cases}$$

Then

$$u_{h,0} = \sum_{i \in \text{Free}} U_i \hat{\phi}_i \quad , \quad u_{h,D} = \sum_{j \in \text{Fixed}} u_{h,Dj} \phi_j$$

where Fixed is the set of nodes on $\partial\Omega_D$, Free is the set of all other nodes.

Inserting $u_h = u_{h,0} + u_{h,D}$ into the weak form

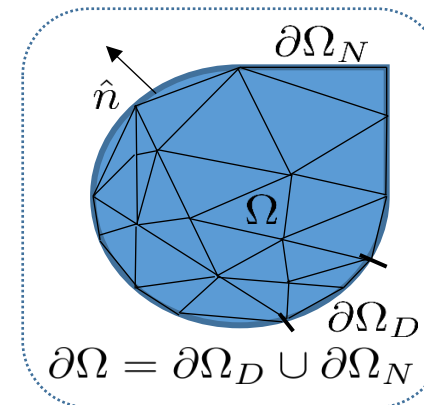
$$\int_{\Omega} \nabla u_h \cdot \nabla v_h \, dV = \int_{\partial\Omega_N} g v_h \, dS + \int_{\Omega} v_h f \, dV$$

the following stiffness matrix and load vector are obtained:

$$\tilde{K}_{ij} = \int_{\Omega} \nabla \hat{\phi}_i \cdot \nabla \hat{\phi}_j \, dV \quad , \quad \tilde{L}_i = \int_{\partial\Omega_N} \hat{\phi}_i g \, dS + \int_{\Omega} f \hat{\phi}_i \, dV - \sum_{j \in \text{Fixed}} u_{h,Dj} \int_{\Omega} \nabla \hat{\phi}_i \cdot \nabla \phi_j \, dV$$

Poisson: $-\nabla^2 u = f$ in Ω
 $u = u_0$ on $\partial\Omega_D$
 $\nabla u \cdot \hat{n} = g$ on $\partial\Omega_N$

$T_h = \{u_h \in P_1 \mid u_h = u_0 \text{ on } \partial\Omega_D\}$
 $\hat{T}_h = \{v_h \in P_1 \mid v_h = 0 \text{ on } \partial\Omega_D\}$



Dirichlet Boundary Conditions

If the free nodes are numbered from 1 to n_f and the fixed nodes are numbered from $n_f + 1$ to n_p , the equation system

$$\tilde{K}_{ij} = \int_{\Omega} \nabla \hat{\phi}_i \cdot \nabla \hat{\phi}_j \, dV \quad , \quad \tilde{L}_i = \int_{\partial\Omega_N} \hat{\phi}_i g \, dS + \int_{\Omega} f \hat{\phi}_i \, dV - \sum_{j \in \text{Fixed}} u_{h,Dj} \int_{\Omega} \nabla \hat{\phi}_i \cdot \nabla \phi_j \, dV$$

can be written as the $n_p \times n_p$ equation system

$$\begin{bmatrix} \tilde{K} & A \\ 0 & I \end{bmatrix} \begin{bmatrix} U_{Free} \\ U_{Fixed} \end{bmatrix} = \begin{bmatrix} L_{Free} \\ u_{h,D} \end{bmatrix}$$

where I is a $(n_p - n_f) \times (n_p - n_f)$ identity matrix and

$$A_{ij} = \int_{\Omega} \nabla \hat{\phi}_i \cdot \nabla \phi_{j+n_f} \, dV \quad ,$$
$$L_{iFree} = \int_{\partial\Omega_N} \hat{\phi}_i g \, dS + \int_{\Omega} f \hat{\phi}_i \, dV$$

for $i \in [1, n_f]$ and $j \in [1, n_p - n_f]$.

The values of the free nodes are obtained by solving

$$\tilde{K}U_{Free} = L_{Free} - AU_{Fixed} = L_{Free} - Au_{h,D}$$

Assembly

$$T_h = \{u_h \in P_1 \mid u_h = u_0 \text{ on } \partial\Omega_D\}$$

$$\hat{T}_h = \{v_h \in P_1 \mid v_h = 0 \text{ on } \partial\Omega_D\}$$

Note that all the $\hat{\phi}_i$'s and the ϕ_j 's are the same *kind* of (hat) functions.

Since the solution is specified at the Dirichlet-nodes, there are no $\hat{\phi}_i$ associated with Dirichlet nodes.

There are thus more ϕ_j basis functions than $\hat{\phi}_i$ basis functions and $\phi_j = \hat{\phi}_j$ for a non-Dirichlet node number j .

We can thus evaluate for all ϕ_i 's insted of the $\hat{\phi}_i$'s to get a $n_p \times n_p$ stiffness matrix and n_p -sized load vector

$$K_{ij} = \int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j \, dV \quad , \quad L_i = \int_{\partial\Omega_N} \phi_i g \, dS + \int_{\Omega} f \phi_i \, dV$$

The integrals can be evaluated efficiently as

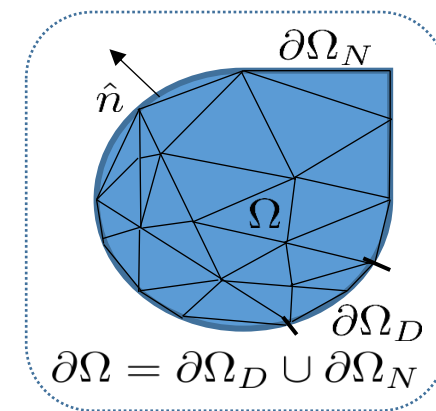
$$K_{ij} = \sum_{k=1}^{n_t} \int_{K_k} \nabla \phi_i \cdot \nabla \phi_j \, dV \quad , \quad L_i = \sum_{e=1}^{n_e} \int_{E_e} \phi_i g \, dS + \sum_{k=1}^{n_t} \int_{K_k} f \phi_i \, dV$$

since the hat functions are only non-zero on triangles containing node i .

K_k is triangle no. k , E_e is edge no. e on $\partial\Omega_N$ and n_e is the number of edges in $\partial\Omega_N$.

The matrices on the previous slide are found as

$$\tilde{K} = K(\text{Free}, \text{Free}) \quad , \quad A = K(\text{Free}, \text{Fixed}).$$



FEM solution process for PDEs

1. Rewrite the PDE into weak form and in the process chose appropriate function spaces for the test and trial functions.
2. Assemble the stiffness matrix and load vector.
3. Apply the boundary conditions.
4. Solve the resulting system of linear equations.
5. Calculate interesting properties from the solution.

FEniCS will make it easy to implement points 2, 3, 4, and 5.

Point 1 can be easy or not-so-easy to obtain using manual labor or a search in books, papers, google, ...

If an energy functional exist for the problem, point 1 can also be obtained using FEniCS. We will see this later...

We will now look at some useful theoretical results!

We will skip some details, but if you are very interested in FEM theory you should do all the derivations yourself 😊

Recap: The L^2 Projection

If we have a function $f \in L^2(\Omega)$ (meaning $\int_{\Omega} |f|^2 d\Omega < \infty$) over the domain Ω , we can approximate the function by a function from $P_1(\Omega)$, where

$$P_1(\Omega) = \left\{ u \mid u(x_1, x_2) = \sum_{i=1}^{n_p} u_i \phi_i(x_1, x_2) , (x_1, x_2) \in M(\Omega) \right\} , \quad \text{and } \phi_i \text{ are hat-functions.}$$

We get interpolation by requiring $u(N_i) = f(N_i) = u_i$ and the function and the approximation (interpolant) matches exactly at the nodal points of the mesh.

Another approximation, $P_h f \in P_1(\Omega)$, can be made by calculating the L^2 projection of f , defined by

$$\int_{\Omega} (f - P_h f) \phi_i dV = 0 \quad , \quad i = 1, \dots, n_p$$

Since $P_h f \in P_1(\Omega)$, then $P_h f = \sum_{j=1}^{n_p} U_j \phi_j$ and

$$\int_{\Omega} \left(f - \sum_{j=1}^{n_p} U_j \phi_j \right) \phi_i dV = 0 \Leftrightarrow \sum_{j=1}^{n_p} U_j \int_{\Omega} \phi_i \phi_j dV = \int_{\Omega} f \phi_i dV \quad , \quad i = 1, \dots, n_p$$

Defining

$$M_{ij} = \int_{\Omega} \phi_i \phi_j dV \text{ (mass matrix)} \quad , \quad L_i = \int_{\Omega} f \phi_i dV \text{ (load vector)}$$

This becomes the $n_p \times n_p$ system of equations $\sum_{j=1}^{n_p} M_{ij} U_j = L_i \quad , \quad i = 1, \dots, n_p$

L^2 Projection

Solving the matrix system, gives us the expansion coefficients, U_i , of the L^2 projection of f on to the function space $P_1(\Omega)$.

The L^2 projection does not necessarily match f at the nodal points of the mesh, instead it minimizes the L^2 error.

Theorem: The L^2 projection, $P_h f$, of f onto $P_1(\Omega)$ satisfies the best-approximation result

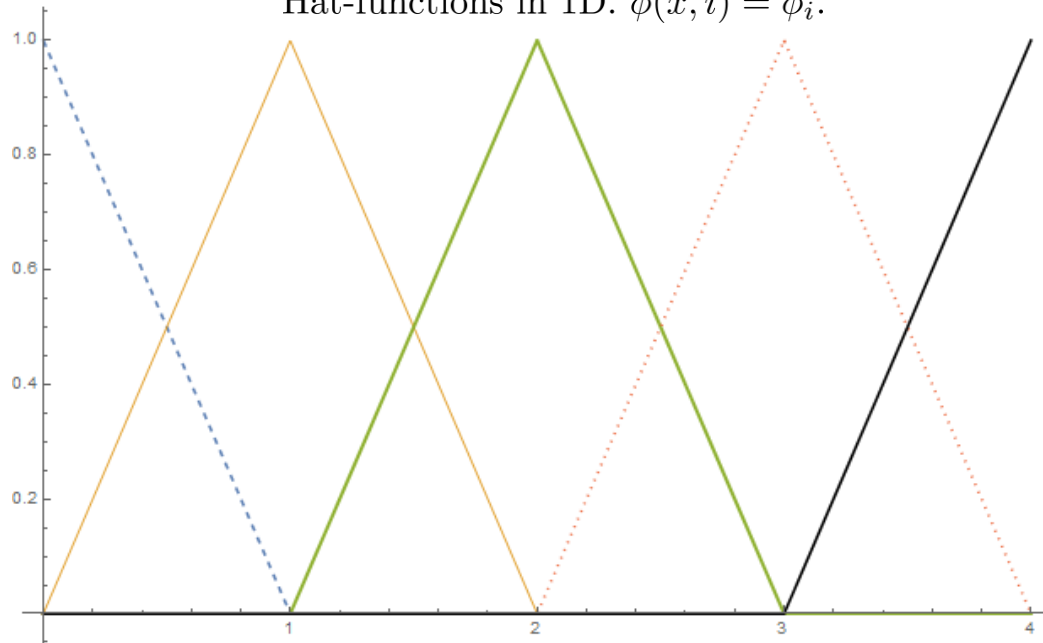
$$\|f - P_h f\|_{L^2(\Omega)} \leq \|f - v\|_{L^2(\Omega)} \quad \forall v \in P_1(\Omega)$$

Proof: Not shown :-)

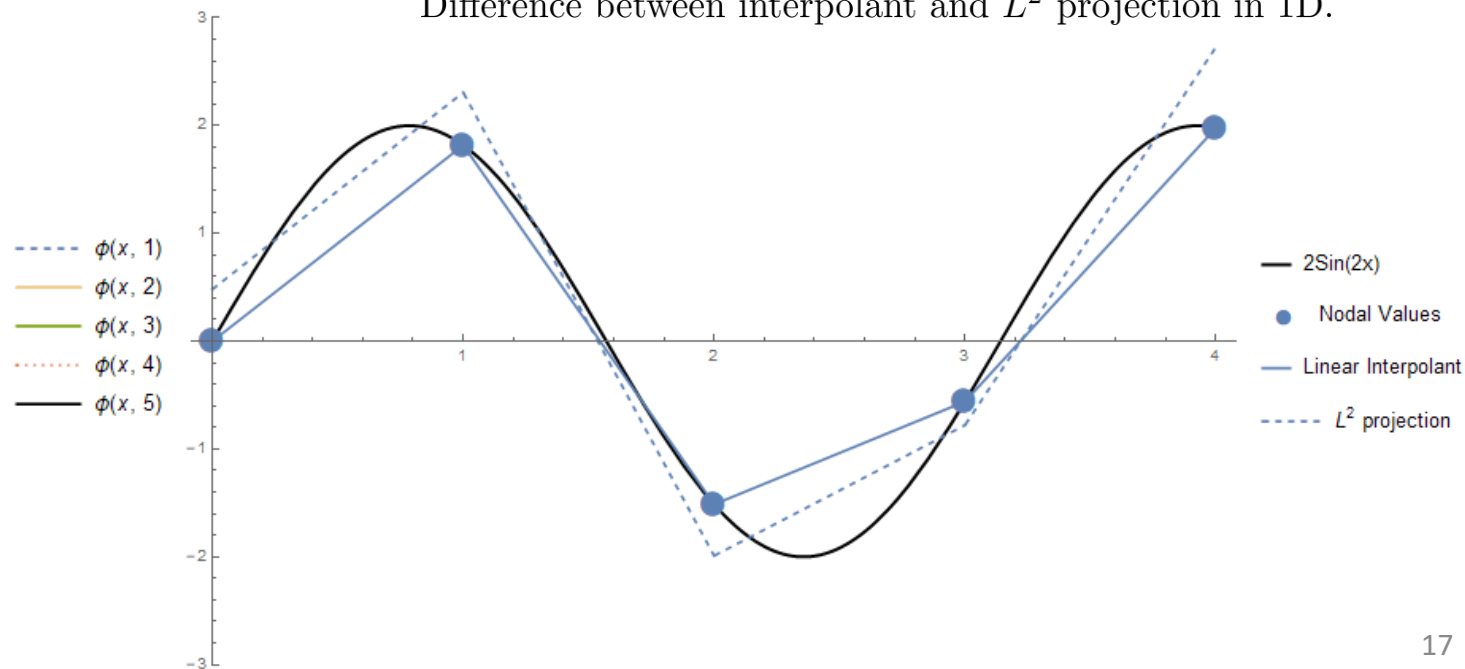
$$\begin{aligned} L^2 \text{ norm:} \\ \|f\|_{L^2} &= \sqrt{\int_{\Omega} |f|^2 dV} \\ L^2 \text{ error:} \\ \|e\|_{L^2} &= \sqrt{\int_{\Omega} |f - P_h f|^2 dV} \end{aligned}$$

The L^2 projection is thus the function in the function space (here $P_1(\Omega)$) with the lowest L^2 error with respect to the original function f , i.e. it is "good overall".

Hat-functions in 1D. $\phi(x, i) = \phi_i$.



Difference between interpolant and L^2 projection in 1D.



Properties of the Finite Element Solution

We focus now on the simplest case

$$\begin{aligned} -\nabla^2 \phi &= f & \text{in } \Omega &, \\ \phi &= 0 & \text{on } \partial\Omega & \end{aligned}$$

New notation:

ϕ is now the unknown function we want to find,
with expansion coefficients ϕ_i .
 ψ_i are the basis functions.

with corresponding weak form problem: Find $\phi \in T_0$ such that

$$\int_{\Omega} \nabla \phi \cdot \nabla v \, dV = \int_{\Omega} f v \, dV \quad , \quad \forall v \in T_0$$

The finite element approximation of this problem is: Find $\phi_h \in T_{h,0}$ such that

$$\int_{\Omega} \nabla \phi_h \cdot \nabla v_h \, dV = \int_{\Omega} f v_h \, dV \quad , \quad \forall v_h \in T_{h,0}$$

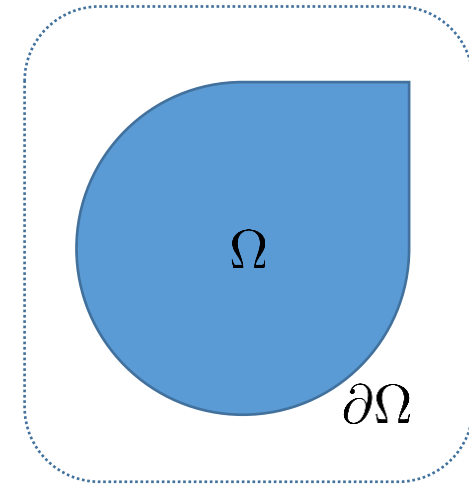
This can also be written as a set of N linear equations with N unknown,

$$\begin{aligned} \sum_{j=1}^N K_{ij} \phi_j &= L_i \quad , \quad \phi_h = \sum_{i=1}^N \phi_i \psi_i \\ K_{ij} &= \int_{\Omega} \nabla \psi_j \cdot \nabla \psi_i \, dV \quad , \quad L_i = \int_{\Omega} f \psi_i \, dV \quad , \quad i, j = 1, 2, \dots, N \end{aligned}$$

T_0 is a function space defined as

$$T_0 = \{v \in T \mid v = 0 \text{ on } \partial\Omega\} \quad , \quad T = \{v \mid \|v\|_{L^2} + \|\nabla v\|_{L^2} < \infty\}$$

and $T_{h,0}$ is a subspace of T_0 with a finite number of basis functions ψ_i , $i = 1, \dots, N$. We assume P_1 functions in the following.



Properties of the Finite Element Solution: Existence and Uniqueness

Theorem: The finite element solution $\phi_h \in T_{h,0}$ to the Poisson problem on the previous slide exist and is unique.

Proof of uniqueness:

By contradiction: Assume there are two different solutions, $\phi_{h,1}$ and $\phi_{h,2}$, both satisfying the finite element problem

$$\int_{\Omega} \nabla \phi_h \cdot \nabla v_h \, dV = \int_{\Omega} v_h \, f \, dV \quad , \quad \forall v_h \in T_{h,0}$$

then

$$\begin{aligned} \int_{\Omega} \nabla \phi_{h,1} \cdot \nabla v_h \, dV &= \int_{\Omega} v_h \, f \, dV \quad , \quad \forall v_h \in T_{h,0} \\ \int_{\Omega} \nabla \phi_{h,2} \cdot \nabla v_h \, dV &= \int_{\Omega} v_h \, f \, dV \quad , \quad \forall v_h \in T_{h,0} \end{aligned}$$

Subtracting the equations:

$$\int_{\Omega} \nabla (\phi_{h,1} - \phi_{h,2}) \cdot \nabla v_h \, dV = 0 \quad , \quad \forall v_h \in T_{h,0}$$

Since this is satisfied for all v in $T_{h,0}$, it must be satisfied for the specific choice $v_h = \phi_{h,1} - \phi_{h,2}$, giving

$$\int_{\Omega} \nabla (\phi_{h,1} - \phi_{h,2}) \cdot \nabla (\phi_{h,1} - \phi_{h,2}) \, dV = \int_{\Omega} |\nabla (\phi_{h,1} - \phi_{h,2})|^2 \, dV = 0$$

This means that $\phi_{h,1} - \phi_{h,2}$ is constant. From the Dirichlet boundary conditions, this constant must be zero, i.e. $\phi_{h,1} - \phi_{h,2} = 0 \Leftrightarrow \phi_{h,1} = \phi_{h,2}$. So the two solutions assumed to be different are the same, contradicting the original statement that the solution is non-unique.

Proof of existence: The solution ϕ_h can be found by solving the N linear equations with N unknown. From the solution of linear equations, such a set of equations have either a unique solution, infinitely many solutions or the equations are inconsistent. Since we know ϕ_h is unique, it follows that it must also exist since we can find it by solving the linear equations.

Properties of the Finite Element Solution: Galerkin Orthogonality

Theorem: The finite element solution ϕ_h satisfies the (Galerkin) orthogonality condition

$$\int_{\Omega} \nabla (\phi - \phi_h) \cdot \nabla v_h \, dV = 0 \quad , \quad \forall v_h \in T_{h,0}$$

where ϕ is the exact solution and ϕ_h is the finite element approximation.

Proof:

ϕ and ϕ_h satisfy

$$\begin{aligned} \int_{\Omega} \nabla \phi \cdot \nabla v \, dV &= \int_{\Omega} v \, f \, dV \quad , \quad \forall v \in T_0 \\ \int_{\Omega} \nabla \phi_h \cdot \nabla v_h \, dV &= \int_{\Omega} v_h \, f \, dV \quad , \quad \forall v_h \in T_{h,0} \end{aligned}$$

Since $T_{h,0} \subset T_0$, we take $v = v_h$ and subtract the equations to get

$$\int_{\Omega} \nabla (\phi - \phi_h) \cdot \nabla v_h \, dV = 0 \quad , \quad \forall v_h \in T_{h,0}$$

which proves the claim.

Galerkin Orthogonality

Let us re-write the Galerkin orthogonality condition as

$$\begin{aligned} 0 &= \int_{\Omega} \nabla (\phi - \phi_h) \cdot \nabla v_h \, dV = \int_{\Omega} (\nabla \phi \cdot \nabla v_h - \nabla \phi_h \cdot \nabla v_h) \, dV \\ &= \int_{\Omega} ((\nabla \cdot (v_h \nabla \phi) - v_h \nabla^2 \phi) - (\nabla \cdot (v_h \nabla \phi_h) - v_h \nabla^2 \phi_h)) \, dV = \int_{\Omega} (-v_h \nabla^2 \phi + v_h \nabla^2 \phi_h) \, dV \end{aligned}$$

where the divergence theorem was used and the $v_h = 0$ on $\partial\Omega$ conditions was used to eliminate the resulting surface terms.

Consider the PDEs

$$-\nabla^2 \phi - f = 0 \Leftrightarrow -\nabla^2 \phi = f \quad \text{and} \quad -\nabla^2 \phi_h - f = -R \Leftrightarrow -\nabla^2 \phi_h = f - R \Leftrightarrow \nabla^2 \phi_h = R - f$$

where R is the residual we get since ϕ_h is an approximation. Inserting into the integrals we get

$$0 = \int_{\Omega} (-v_h \nabla^2 \phi + v_h \nabla^2 \phi_h) \, dV = \int_{\Omega} (v_h f + v_h (R - f)) \, dV = \int_{\Omega} v_h R \, dV$$

Note that this is true for all $v_h \in T_{h,0}$, so the residual is orthogonal to all functions in the function space $T_{h,0}$. This means that we can not make the residual smaller within the function space $T_{h,0} \subset T_0$.

Choosing $v_h = \phi_i$, we get

$$\int_{\Omega} R \phi_i \, dV = 0 \quad , \quad i = 1, \dots, n_p$$

which shows that FEM is also a weighted residual method.

Product rule

$$\nabla \cdot (v \vec{F}) = \nabla v \cdot \vec{F} + v \nabla \cdot \vec{F} \Leftrightarrow$$

$$\nabla v \cdot \vec{F} = \nabla \cdot (v \vec{F}) - v \nabla \cdot \vec{F}$$

Divergence theorem

$$\int_{\Omega} \nabla \cdot \vec{F} \, dV = \oint_{\partial\Omega} \vec{F} \cdot \hat{n} \, dS$$

Galerkin Orthogonality

Choosing $v_h = \phi_i$, we get

$$\int_{\Omega} R \phi_i \, dV = 0 \quad , \quad i = 1, \dots, n_p$$

which shows that FEM is also a weighted residual method.

Since $R = R_{exact} - R_{approx}$, where $R_{exact} = 0$ is the exact residual, then

$$-\int_{\Omega} R \phi_i \, dV = -\int_{\Omega} (R_{exact} - R_{approx}) \phi_i \, dV = 0 \quad , \quad i = 1, \dots, n_p$$

and we can view this as the L^2 projection of the residual onto the space $T_{0,h}$.

The residual corresponding to the solution ϕ_h will be "overall small" over the geometry.

L^2 projection:

$$\int_{\Omega} (f - P_h f) \phi_i \, dV = 0 \quad , \quad i = 1, \dots, n_p$$

Properties of the Finite Element Solution: Best Approximation

Theorem: The finite element solution ϕ_h satisfies the "best approximation"-condition

$$\sqrt{\int_{\Omega} |\nabla (\phi - \phi_h)|^2 dV} = |||\phi - \phi_h||| \leq |||\phi - v_h||| = \sqrt{\int_{\Omega} |\nabla (\phi - v_h)|^2 dV} \quad , \quad \forall v_h \in T_{h,0}$$

$|||v||| = \|\nabla v\|_{L^2}$ is named the energy norm and $\phi - \phi_h$ is the error.

Proof:

Let $v_h \in T_{h,0}$, then

Galerkin orthogonality:

$$\int_{\Omega} \nabla (\phi - \phi_h) \cdot \nabla v_h dV = 0 \quad , \quad \forall v_h \in T_{h,0}$$

$$\begin{aligned} |||\phi - \phi_h|||^2 &= \int_{\Omega} |\nabla (\phi - \phi_h)|^2 dV = \int_{\Omega} \nabla (\phi - \phi_h) \cdot \nabla (\phi - \phi_h) dV \\ &= \int_{\Omega} \nabla (\phi - \phi_h) \cdot \nabla (\phi - v_h) dV + \int_{\Omega} \nabla (\phi - \phi_h) \cdot \nabla (v_h - \phi_h) dV \\ &= \int_{\Omega} \nabla (\phi - \phi_h) \cdot \nabla (\phi - v_h) dV \end{aligned}$$

$$\phi - \phi_h = (\phi - v_h) + (v_h - \phi_h)$$

where the last equality follow from Galerkin orthogonality, since $(v_h - \phi_h) \in T_{h,0}$.

The L^2 -norms satisfy the Cauchy-Schwarz inequality

$$\int_{\Omega} vw dV \leq ||v|| ||w|| \quad (\text{think } |\vec{v} \cdot \vec{w}| = |\vec{v}||\vec{w}| |\cos \phi| \leq |\vec{v}||\vec{w}|)$$

so

$$|||\phi - \phi_h|||^2 = \int_{\Omega} \nabla (\phi - \phi_h) \cdot \nabla (\phi - v_h) dV \leq |||\phi - \phi_h||| |||\phi - v_h|||$$

Dividing by $|||\phi - \phi_h|||$ concludes the proof.

Remember the properties of the linear interpolant: Error Estimate

Theorem: Given a continuous function f on a triangle K with nodes N_i , $i=1,2,3$, the linear interpolant of f over K is $\pi f \in P_1(K)$ given by

$$\pi f = \sum_{i=1}^3 f(N_i) \phi_i.$$

The interpolant πf satisfies the error estimates

$$\|f - \pi f\|_{L^2(K)} \leq Ch_K^2 \|D^2 f\|_{L^2(K)} \quad , \quad \|D(f - \pi f)\|_{L^2(K)} \leq Ch_K \|D^2 f\|_{L^2(K)}$$

where h_K is the longest edge length of the triangle and

$$Df = \sqrt{\left|\frac{\partial f}{\partial x_1}\right|^2 + \left|\frac{\partial f}{\partial x_2}\right|^2} \quad , \quad D^2 f = \sqrt{\left|\frac{\partial^2 f}{\partial x_1^2}\right|^2 + 2\left|\frac{\partial^2 f}{\partial x_1 \partial x_2}\right|^2 + \left|\frac{\partial^2 f}{\partial x_2^2}\right|^2}.$$

Proof: Not shown... :-)

Remember that C is large if the triangle is "stretched".

Properties of the Finite Element Solution: Error Estimate

Theorem: The finite element solution ϕ_h satisfies the error estimate

$$|||\phi - \phi_h|||^2 \leq \sum_{k \in \kappa} C_k h_k^2 \|D^2 \phi\|_{L^2(k)}^2$$

C_k will be large if the triangle is very "stretched".

Proof:

From the "best approximation"-theorem

$$|||\phi - \phi_h|||^2 \leq |||\phi - v_h|||^2 \quad \forall v_h \in T_{h,0}$$

Choosing $v_h = \pi\phi \in T_{h,0}$, the linear interpolant to the solution ϕ , we get

$$|||\phi - \phi_h|||^2 \leq |||\phi - \pi\phi|||^2 = \sum_{k \in \kappa} \|D(\phi - \pi\phi)\|_{L^2(k)}^2 \leq \sum_{k \in \kappa} C_k h_K^2 \|D^2 \phi\|_{L^2(k)}^2$$

And the claim is shown. Note also that we can also write $|||\phi - \phi_h||| \leq Ch \|D^2 \phi\|_{L^2(\Omega)}$.

Properties of the Finite Element Solution: Error Estimate

The error estimate

$$|||\phi - \phi_h||| \leq Ch ||D^2\phi||_{L^2(\Omega)}$$

tells us that when the mesh size h is decreased, so is (the gradient of) the error.

Also, if the true solution wiggles a lot, then $D^2\phi$ is large and the error is large.

C will be large if any of the triangles are very "stretched", making again the error large.

We must generally decrease h if we want to bring the error down.

The estimate can be converted to an estimate directly on the error, by introducing a dual problem. This result in:

Theorem: The finite element solution ϕ_h to the Poisson problem satisfies the error estimate

$$||\phi - \phi_h||_{L^2(\Omega)} \leq Ch^2 ||D^2\phi||_{L^2(\Omega)}$$

Proof: Not shown... :-)

This shows a 2nd order convergence in the mesh-size h , when using linear basis (shape) functions.

We can see how the error behaves, but we can not really estimate the actual error from this expression, as it depends on the exact solution, ϕ .

This is known as an "a-priori error estimate".

More generally, for polynomial basis functions of order p , the error is expected to decrease as h^{p+1} .

Properties of the Finite Element Solution: A Posteriori Error Estimate

If we can estimate the error of the finite element solution, we can use this to increase the accuracy.

The error is not the same at all points in the mesh, if e.g. the true solution is almost constant in one part of the domain but highly wiggling in another, the a-priori errors tells us that the error will be small in the almost constant region and high in the highly wiggling region.

If we have this knowledge, we can refine the mesh in the highly varying part and leave it coarse in the almost constant part of the domain. But the a-priori error estimate does not allow us to compute the error estimate!

We need a "posteriori error estimate".

Theorem: The finite element solution to the Poisson problem satisfies the error estimate

$$|||\phi - \phi_h|||^2 \leq C \sum_{k \in \kappa} \eta_k^2(\phi_h) \quad , \quad \eta_k(\phi_h) = h_k ||f + \Delta \phi_h||_{L^2(k)} + \frac{1}{2} \sqrt{h_k} ||[n \cdot \nabla \phi_h]||_{L^2(\partial k \setminus \partial \Omega)}$$

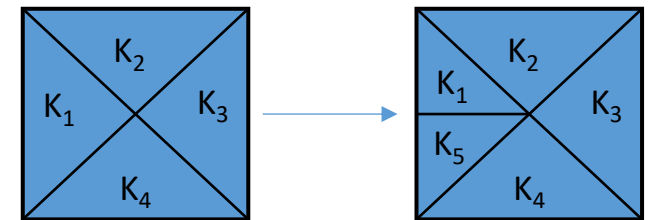
where $[n \cdot \nabla \phi_h]$ is the jump in the normal derivative of ϕ_h when moving from one triangle to another. And $\Delta \phi_h = \nabla^2 \phi_h = 0$ since our solution is assumed to use linear shape functions.

Proof: Not shown... :-)

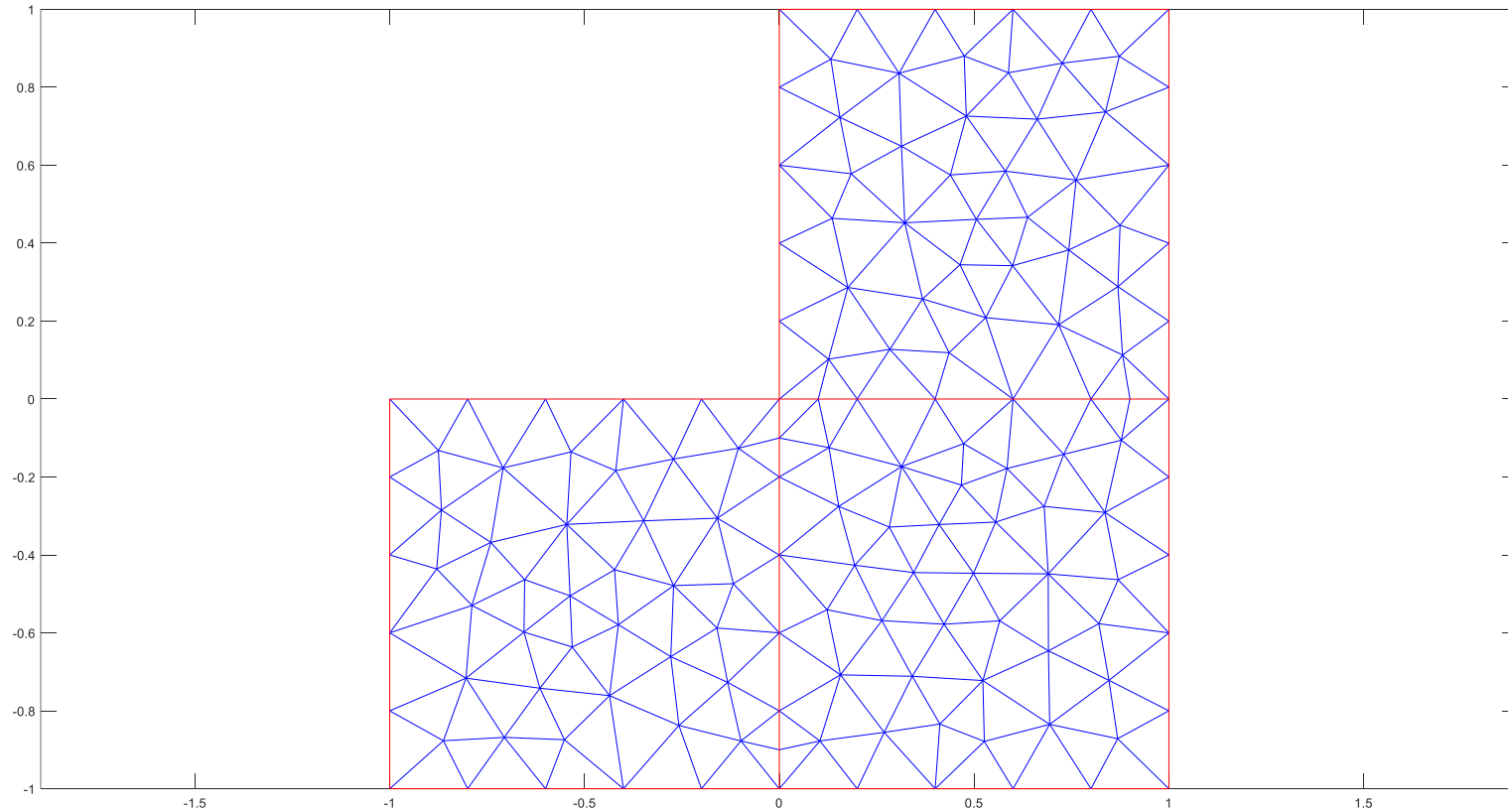
We can now calculate the error estimate in each element from the finite element solution.

This can be used to find the elements with the largest error and divide them into smaller elements.

The new triangles generated should always be kept as regular as possible, otherwise we might introduce additional errors into the solution.



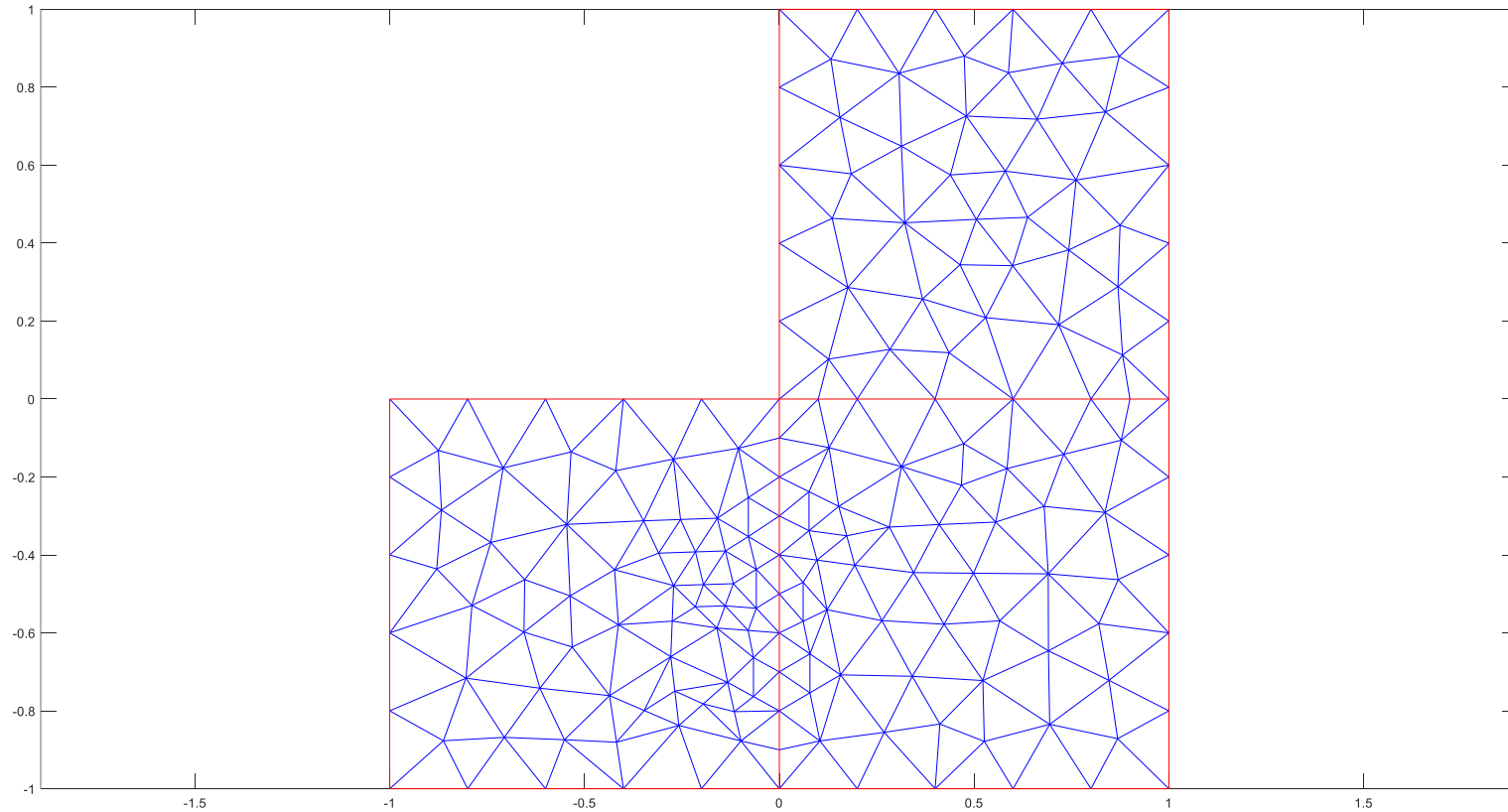
Adaptive Mesh Refinement Idea



Solve problem on 1st Mesh.

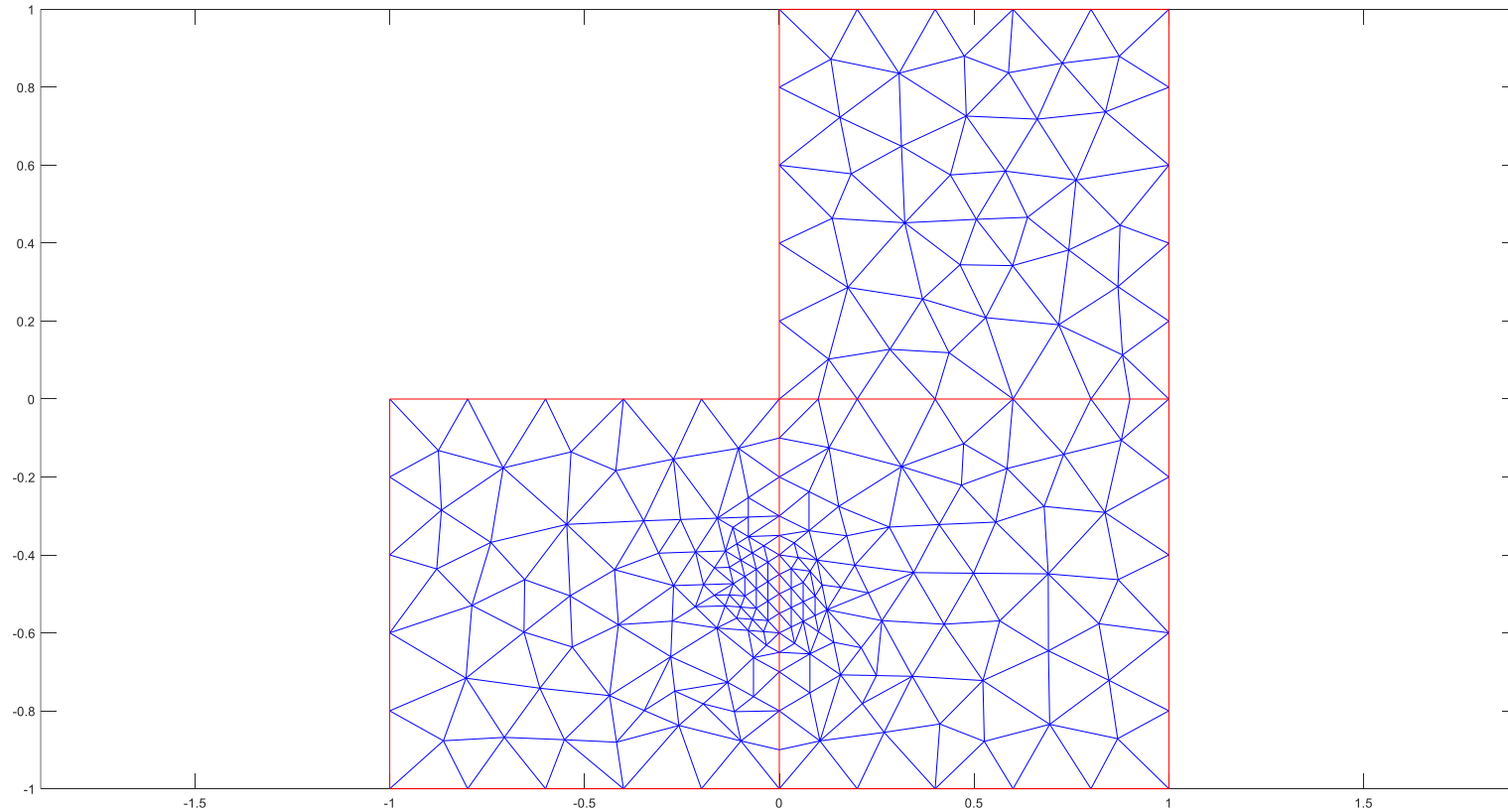
Then calculate element errors and refine where error is large.

Adaptive Mesh Refinement Idea



Solve problem on 2nd Mesh.
Then calculate element errors and refine where error is large.

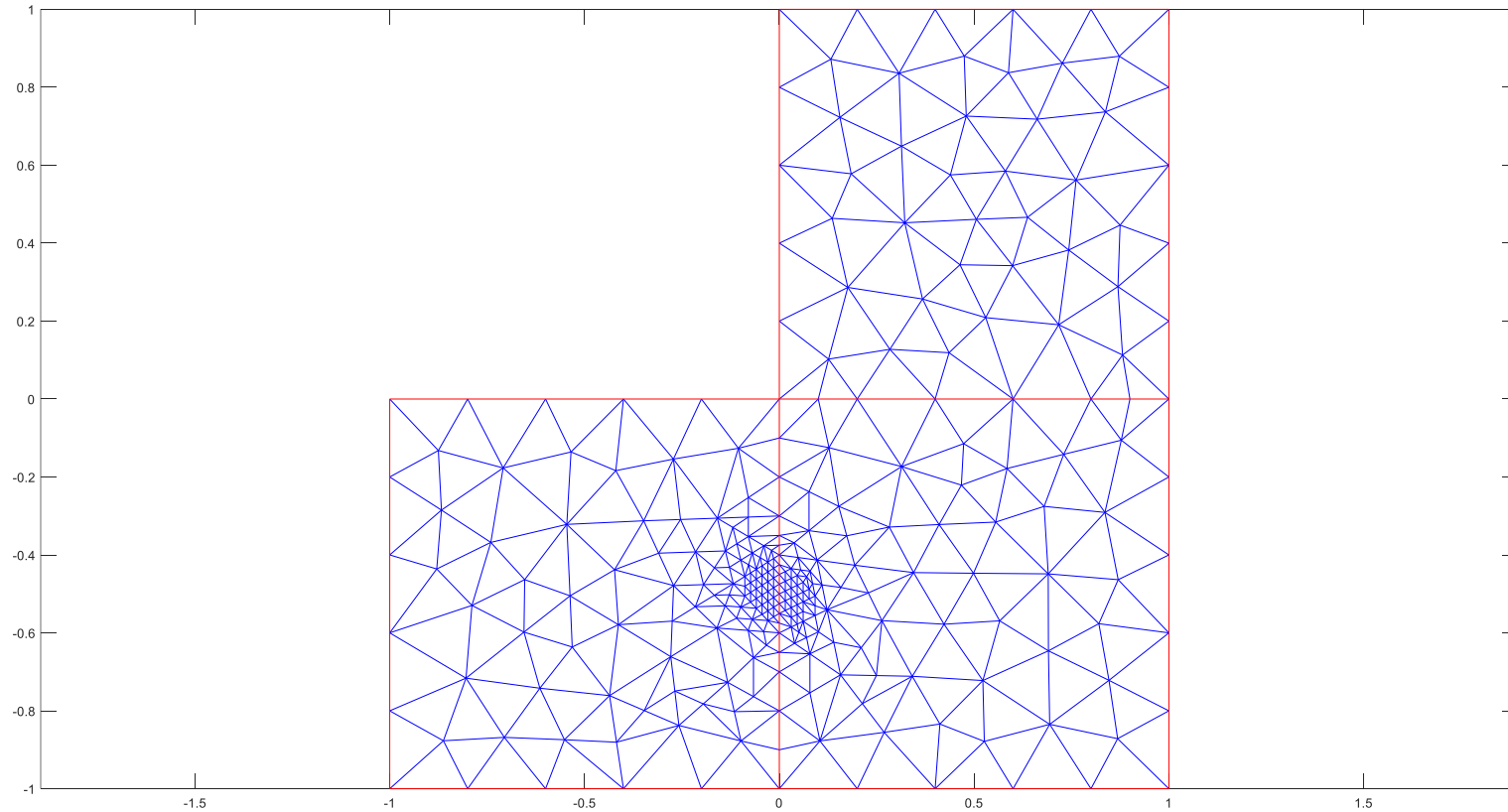
Adaptive Mesh Refinement Idea



Solve problem on 3rd Mesh.

Then calculate element errors and refine where error is large.

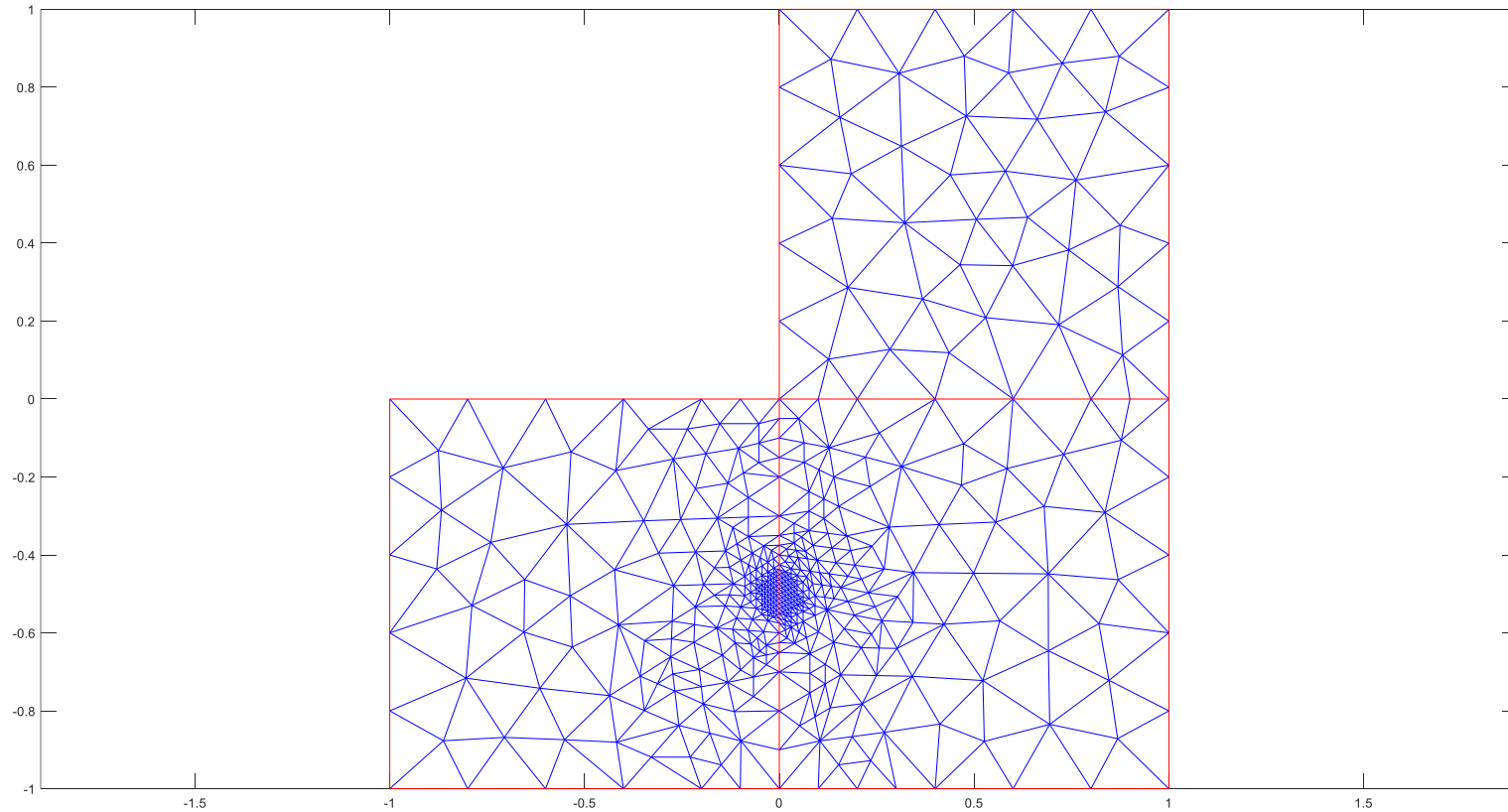
Adaptive Mesh Refinement Idea



Solve problem on 4th Mesh.

Then calculate element errors and refine where error is large.

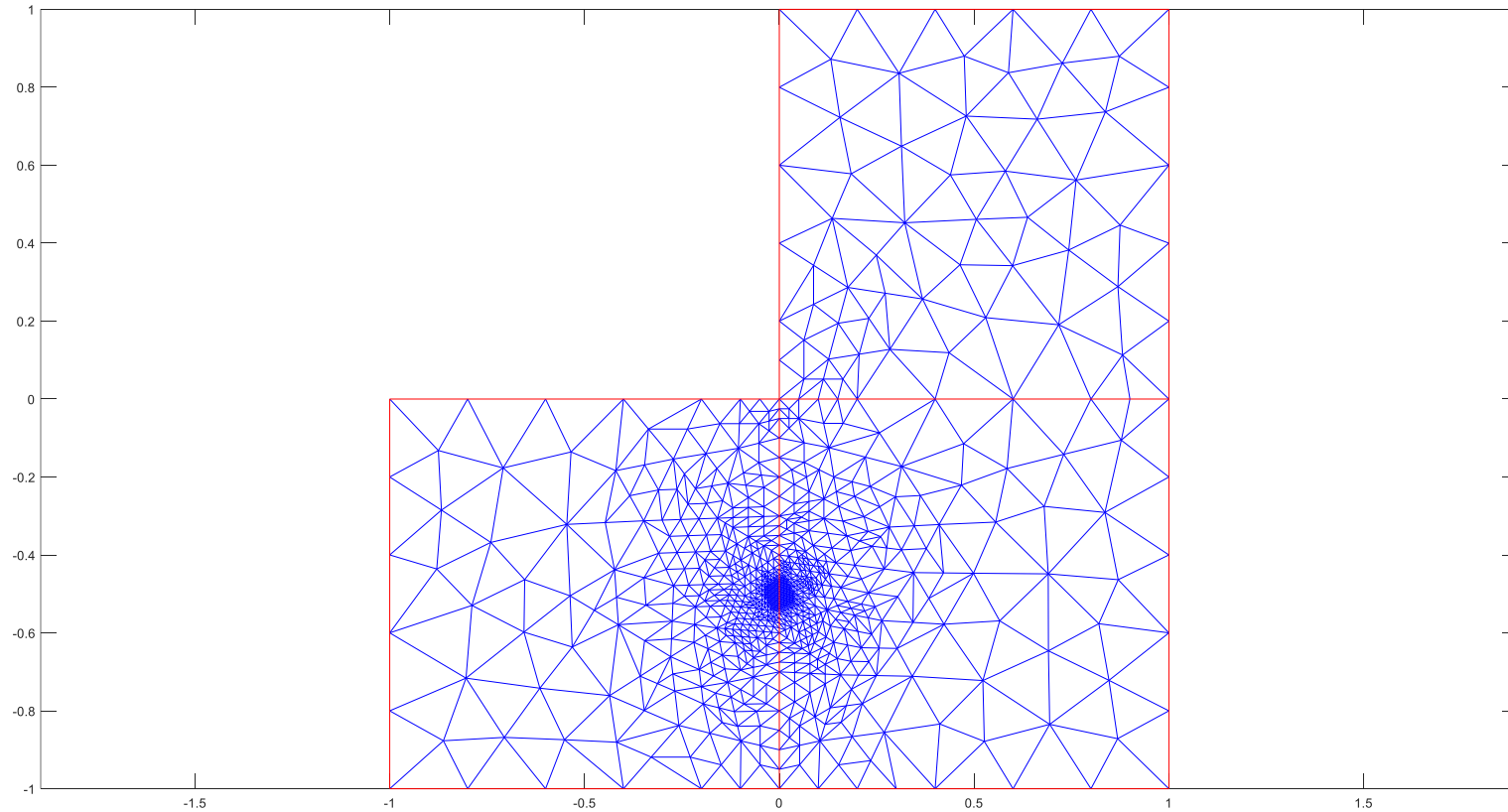
Adaptive Mesh Refinement Idea



Solve problem on 5th Mesh.

Then calculate element errors and refine where error is large.

Adaptive Mesh Refinement Idea



Solve problem on 6th Mesh.
Then calculate element errors.
Now the error is so small that we stop.

PS: We will not use adaptive mesh refinement in this course, but it is a neat application of the theoretical FEM results 😊 33